Let's talk about... computing fundamentals workbook

COURSE OVERVIEW

This course explores some key vocabulary in computing and when using the BBC micro:bit. It aims to help you make connections to what you already know and empower you to feel more confident when introducing and using these terms in your lessons.

USEFUL DEFINITIONS WITH EXAMPLES

The 'Let's talk about computing fundamentals' course breaks down the meaning of many key computing terms. We have collated the main definitions below, with some examples, for your reference.

VOCABULARY	DEFINITION	EXAMPLE
Algorithms	A set of step-by-step instructions.	Instructions for getting dressed – these need to be followed in a specific order or you could put socks on over shoes!
Sequence	The specific order of instructions.	
Decomposition	Breaking down a problem into smaller parts.	When tasked with planning a party, the task can be broken into smaller tasks such as booking a venue, sending invitations and ordering food.
Abstraction	Identifying and focusing on only the important parts needed to solve the problem.	Abstracting a human face into an emoji, only the eyes and smile are included.
Pattern recognition	Spotting similarities and repetition to improve the efficiency of our instructions.	Instructions for a dance including a pattern could be written as follows: **Repeat 4 times** Move forward Move back Move left Move right Writing it this way is more efficient than writing out each step 4 times.
Loops	Allow you to repeat sets of instructions without having to write them out multiple times.	

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VOCABULARY	DEFINITION	EXAMPLE
Selection	Making different things happen based on different conditions.	Selection adds in some choice in a scenario like checking if it is raining and then deciding if you need an umbrella or not. It adds in more than one path.
Variables	A container for storing data which can be accessed and updated while a program is running.	Top scores recorded on computer games are an example of a variable, as are personalised log-in details.
Debugging	Finding and fixing errors with code.	Possible errors could be logic based (testing for being too hot when meant to test for being too cold), sequence (steps in the wrong order) or spelling mistakes.

USING THIS COURSE IN YOUR CLASSROOM

As well as recapping what the key computing terms mean, we have included some fun ways to explore these terms in your classroom and highlighted how these terms connect with things you and your learners already know.

What you need

Resources:

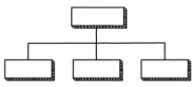
- **Activity plans** These can be downloaded from the course materials page as a zip file and contain unplugged activities to explore each computing concept.
    - Drawing instructions activity plan – exploring algorithms
    - Planning a party activity plan – exploring decomposition
    - Movie in 3 words activity plan – exploring abstraction
    - Movement instructions activity plan – exploring pattern recognition & loops
    - Musical chairs activity plan – exploring selection
    - Human step counter activity plan – exploring variables
    - Everyday instructions activity plan – exploring debugging

- **Projects mentioned in this course**
    - Beating Heart (https://mbit.io/pd-heart-code)
    - Emotion Badge (https://mbit.io/pd-badge-code)
    - Night Light (https://mbit.io/pd-code-nightlight)
    - Step Counter (https://mbit.io/pd-steps)

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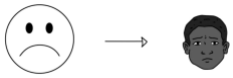
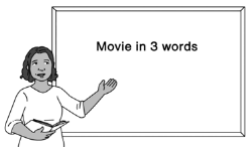
Activity breakdown:

ALGORITHMS & SEQUENCE IDEAS	
	**01** **Introduce** the term algorithm and practice pronouncing it. Ask if anyone knows what it means and share the definition: a set of step-by-step instructions. Ask the class for examples of when they have already followed a specific set of instructions today, for example, washing their hands.
	**02** **Explain** that algorithms have 2 key elements: sequence and precision. Share the definition for sequence: the specific order of instructions. Ask the learners if they think the order of instructions matter? Discuss how the order is important using the example of putting on your clothes each day.
	**03** **Introduce** the term precision and share the definition: the clarity and accuracy of the instructions. Use the *Drawing instructions activity plan* to explore the importance of using precise language.
	**04** **Discuss** how algorithms are all around us in everyday life and school subjects (for example, recipes or Mathematics). Ask the class to spot examples of algorithms around them over the next few days and you can make a note of any you spot too.

DECOMPOSITION IDEAS	
	**05** **Share** the definition of decomposition: breaking down a problem into smaller parts. **Challenge** your learners to start organising a party using the *Planning a party activity plan*.
	**06** **Highlight** why decomposition is useful in computing and how we already use it in everyday problem solving. Ask learners to think about where they use decomposition to break down things into smaller parts. Writing stories, learning new skills, facing a problem etc. What are the benefits of breaking it down?

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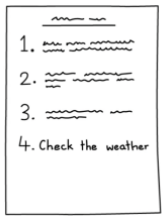
ABSTRACTION IDEAS

	**07** **Share** the definition of abstraction: identifying and focusing on only the important parts needed to solve the problem. **Highlight** how it is a useful tool for problem solving and it is a name for something that we all regularly do without noticing.
	**08** **Share** emojis as a good example of abstraction. Share some face emojis and ask what facial features are / are not included? Ask learners to pair up and spot which of their partner's facial features move/change as they express different emotions. Use a happy face emoji as an example of abstraction: when people smile their mouth and eyes change shape, but not their ears or hair.
	**09** **Use** the *Movie in 3 words activity instructions* to practice abstraction with your class.
	**10** **Discuss** why abstraction helps with problem solving and helps limit the 'noise of extra information', leading to better solutions. Encourage them to think about where they might already abstract down to the important information such as writing a book report.

PATTERN RECOGNITION & LOOPS IDEAS

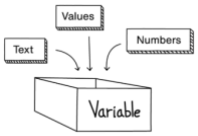
	**11** **Introduce** the definition for pattern recognition: spotting similarities and repetition to improve the efficiency of our instructions. **Discuss** how pattern spotting is a skill they master early in life and display some visual patterns or music notes with patterns asking the class to predict the next item in the pattern. Ask the class to spot some repeating patterns in songs they are singing in school.
	**12** **Challenge** your class to take part in the Movement instructions activity as described in the *Movement instructions activity plan*. This activity allows learners to create an algorithm for a movement, identify repetition in the algorithm and use loops to make it more efficient.

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SELECTION IDEAS	
	**13** **Share** the selection definition: making different things happen based on different conditions. Give examples of a 'condition' such as changes in temperature or a button push.
	**14** **Give the example** of writing an algorithm for leaving the house in the morning. Add in a step for checking the weather. This leads to two selection pathways: • if it is raining, take an umbrella • If it is not raining, leave the umbrella at home Instead of a linear algorithm, selection allows us to be more responsive by adding in some choice/decision-making. This allows us to include more than one path.
	**15** **Highlight** how selection can be written as pseudocode using IF and ELSE statements or in a flow chart. **IF** the weather is wet – take umbrella **ELSE** – leave umbrella at home **Discuss** how selection can be trigged by inputs such as pressing a button or detecting light levels. Ask the class to come up with a list of systems that they think have 'decision points' in them. Some examples are automatic doors or smoke alarms.
	**16** **Challenge** your class to think about selection using the *Musical chairs activity plan*. This activity allows learners to use IF and ELSE statements and uses a well-known game to explore selection pathways.
	**17** **Summarise** by looking at how selection is used in other areas such as thinking about how decisions lead to different consequences.

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VARIABLES IDEAS



18 **Share** the definition of a variable: a container for storing data which can be accessed and updated while a program is running.

Ask learners to think of examples of where variables are used in everyday life such as smart watches, computer games or logging in safely to an online bank account.



19 **Challenge** your learners to explore variables through using the *Human step counter activity plan*. This will allow learners to physically model a variable to see how it works.

DEBUGGING IDEAS

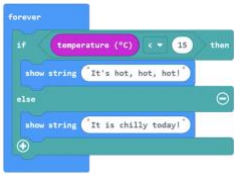


20 **Share** the definition of debugging: finding and fixing errors with code.

Give learners the background to how the term ‘de-bugged’ came about: In 1947 a moth became trapped in a mechanical switch, stopping it from turning on. The person who removed the moth said they had ‘de-bugged’ the machine.



21 **Highlight** that when making computer programs, the problems can be subtle and difficult to spot. Sometimes all the components are correct but perhaps the order is slightly wrong. A good way to demonstrate this is to take an activity that the class are familiar with and mix up the order. Use the *Everyday instructions activity plan* to get them spotting and fixing bugs/errors.



22 **Share** an example of some code with a ‘bug’ and see if the class can spot the error.

Encourage learners to read the code aloud to see if they can spot an error.

Hint – here they need to consider the < symbol (an example of how errors can be subtle).

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AFTER THIS COURSE

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